INTRODUCTION TO APPLICATION

I submitted a K99/R00 application entitled “Statistical Advances in CRISPR Profiling and Screening Analysis” from Harvard School of Public Health (HSPH) on July 12, 2025. The primary mentor on this grant was Dr. Xihong Lin (HSPH), and the co-mentors were Dr. Daniel Bauer (Boston Children’s Hospital; BCH), Dr. Luca Pinello (Massachusetts General Hospital), and Dr. Danwei Huangfu (Memorial Sloan Kettering). Due to funding challenges at HSPH, I transitioned from Dr. Lin’s Lab at HSPH to Dr. Bauer’s Lab at BCH in July of 2025. Dr. Bauer is now my primary mentor, and Dr. Lin is now my co-mentor. Dr. Luca Pinello and Dr. Danwei Huangfu remain co-mentors. Thus, my mentorship team remains the same. Additionally, my science (i.e., Specific Aims, Strategy, and Bibliography) remains the same. BCH and HSPH are allied institutions with shared resources (e.g., computing clusters and postdoc courses), and so my environment remains very similar. All mentors remain enthusiastic about my proposal and are committed to facilitating my transition to an independent faculty position.

I have updated the following components of the application to reflect my change in institution.

1. Mentor letter from Dr. Bauer. Dr. Bauer’s mentor letter now reflects his role as primary mentor.
2. Mentor letter from Dr. Lin. Dr. Lin’s mentor letter now reflects her role as co-mentor.
3. Institutional support statement. An institutional support statement from Dr. David A Williams, chair of the Division of Hematology and Oncology at BCH, has been added.
4. Candidate’s background, career goals and objectives, and training plan. The training plan statement has been updated to reflect Dr. Bauer’s role as primary mentor and Dr. Lin’s role as co-mentor.
5. Responsible conduct of research (RCR). The RCR statement now describes the RCR course offered by BCH, which I plan to take in the fall of 2026.
6. Facilities and other resources. The facilities statement has been updated to include a description of the Harvard Medical School cluster, to which I have access as a postdoctoral researcher at BCH.
7. Environment. The environment statement has been updated to reflect my new environment at BCH.
8. Budget. The budget has been updated to reflect the BCH pay structure.

Additionally, I have obtained a relinquishment letter from HSPH and an SF 424 form from BCH.

Dana-Farber Cancer Institute | Boston Children's | HARVARD MEDICAL SCHOOL TEACHING HOSPITAL

Dana-Farber/Boston Children's Cancer and Blood Disorders Center

Daniel E. Bauer, MD, PhD
Staff Physician • Principal Investigator • Gene Therapy Program Director, Pediatric Hematology/Oncology, Boston Children's Hospital and Dana-Farber Cancer Institute | Donald S. Fredrickson, MD Associate Professor of Pediatrics, Harvard Medical School | Principal Faculty, Harvard Stem Cell Institute | Associate Member, Broad Institute
Karp 8215, 1 Blackfan Circle, Boston MA 02115, t 617-919-2507, f 617-730-0934, daniel.bauer@childrens.harvard.edu, bauerlab.org

October 3rd, 2025

Dear K99/R00 Pathway to Independence Award Committee,

I am delighted to **support as primary mentor the application of Dr. Timothy (Tim) Barry for the NIH K99/R00 Pathway to Independence award**. Tim is a highly motivated early-career scientist brimming with ideas. The statistical methods that he proposes to develop will help resolve critical unmet needs in the analysis of CRISPR perturbation data. His grant aims are ambitious but entirely achievable given his comprehensive training plan and strong background in statistical and computational genomics.

Background. During his productive PhD, Tim focused on developing methods for single-cell CRISPR screen (perturb-seq) analysis. Perturb-seq combines single-cell sequencing with CRISPR genome editing to survey the impact of genetic perturbations on individual cells. Across a series of three first-authored papers published in leading biostatistics and computational genomics journals, Tim developed SCEPTRE, a toolkit for statistically rigorous and massively scalable perturb-seq data analysis. SCEPTRE has rapidly become the standard tool for perturb-seq analysis in the functional genomics field. E.g., SCEPTRE is the first-ever third-party perturb-seq analysis tool endorsed by 10x Genomics (the leading commercial supplier of scRNA-seq kits), has been incorporated into the drug discovery pipeline of CAMP4, a biotech startup, and is widely used by leading functional genomics labs around the world. Tim wrote a compressive e-book – "Hands on single-cell CRISPR screen analysis" (timothy-barry.github.io/sceptre-book/) – guiding users through the entire process of analyzing their perturb-seq data, which has accelerated uptake of the tool by the community. Tim's work has been recognized by the American Statistical Association (Mihaela Serban Memorial Award) and the American Society of Human Genetics (Reviewers' Choice Award) and has garnered hundreds of citations. Tim has collaborated extensively with biologists, fostering interdisciplinary communication and collaboration skills that will facilitate his transition into an independent investigator position. Tim has assisted labs in applying SCEPTRE to study cardiovascular disease, autism spectrum disorder, and blood cell traits, resulting in co-authorship on several publications. Moreover, Tim frequently consults with genomics labs over Zoom, Github, and email to provide guidance on various aspects of perturb-seq experimental design and analysis. Tim helped incorporate SCEPTRE into the official CRISPR screen pipeline for the Impact of Genomic Variation on Function (IGVF) consortium, an NIH-funded program that seeks to understand the biological function of disease-associated variants.

Tim joined the lab of Dr. Xihong Lin at the Harvard School of Public Health (HSPH) in January of 2024. He worked on a statistical methodology project related to differential analysis, which entails comparing the distribution of count data across conditions while adjusting for covariates. Differential analysis is a fundamental statistical task in the analysis of many kinds of genomics data, including single-cell RNA sequencing, bulk RNA sequencing, and CRISPR screen data. Tim proposed the "permuted score test," a novel method for robust differential analysis based on the technique of permuting score test statistics. The permuted score test provably overcomes limitations of existing approaches for differential analysis, reducing false positives and improving power. Tim preprinted a manuscript proposing the permuted score test to arXiv and plans to submit this manuscript to the Journal of the American Statistical Association soon. The permuted score test will play a key role in *Aim 1* of Tim's K99/R00 research (alongside a host of new Bayesian and robust inference methods).

Current work. I met Tim at the Impact of Genomic Variation on Function (IGVF) consortium in Seattle, WA in 2024, where he impressed me with his work on CRISPR screen analysis. I served as a mentor on Tim's initial K99/R00 application, which he submitted from the Harvard School of Public Health in June of 2025. After unexpected funding challenges at Harvard, Tim joined my lab at Boston Children's Hospital (BCH) in July of 2025. Tim has updated his K99/R00 materials to reflect this change in institution, and I am now primary mentor on the application. Tim currently is extending his expertise in CRISPR data analysis towards approaches needed for CRISPR therapeutic development, developing statistical and computational tools to assess the efficacy and safety of

therapeutic CRISPR systems. Tim's K99/R00 proposes to develop innovative statistical methods that will resolve critical unmet needs in the CRISPR therapeutic and screening space. In *Aim 1*, Tim proposes to develop CRISPR-DE, a comprehensive toolkit for estimating editing frequencies and testing for differential editing in CRISPR profiling experiments. Next, in *Aims 2* and *3*, Tim will develop *DML-Perturb* and *IV-Perturb*, methods for accurate, interpretable, and causally rigorous analysis of perturbation screen data through novel deep learning and instrumental variables frameworks. Ultimately, Tim's R00/K99 proposal will yield a cohesive suite of statistical methods for CRISPR profiling and screening analysis, accelerating therapeutic development and biological discovery. I expect Tim to be well positioned to start his own lab in the R00 phase of this award.

Training plan and environment. My lab studies the genetic determinants of blood cell ontogeny and disease. We identified the key erythroid enhancer of *BCL11A* and discovered the guide RNA used in the first-ever FDA-approved gene editing therapy (brand name Casgevy, for sickle cell disease and b-thalassemia). More recently, my lab has optimized therapeutic base and prime editors in hematopoietic stem cells, developed computational tools for characterizing the editing activity of CRISPR systems, and pioneered CRISPR screening strategies for understanding blood cell phenotypes. Since starting my lab 11 years ago, seven of my former students and fellows have gone on to faculty positions directing independent academic research labs and others are full time scientists working in biopharma.

In my capacity as primary mentor, I will work closely with Tim to help him accomplish his training goals and ultimately achieve independence. I will continue to meet with Tim on a biweekly basis to provide feedback on his research, point him to key literature, and direct him to relevant colloquia and career development opportunities. Additionally, I will provide direct, hands-on guidance on all aspects of CRISPR biology and technology relevant to his research, including CRISPR editing mechanisms (e.g., sequence recognition, DNA repair pathways, epigenetic remodeling, etc.), CRISPR screen design and analysis (e.g., pooled and arrayed screens, tiling screens, molecular phenotyping, etc.), and off-target analysis (e.g., amplicon sequencing, DNA tagging sequencing, etc.). Tim will present in my lab meeting quarterly, where he will receive feedback on his work. As Tim prepares for the academic job market, I will coach Tim on writing a research statement, giving a job talk, interviewing with faculty members, and negotiating a start-up package. Finally, once Tim secures an independent position, I will provide guidance on setting lab culture and applying for grants. Tim's comentorship team comprises Dr. Xihong Lin, Dr. Luca Pinello, and Dr. Danwei Huangfu; Tim will meet with Dr. Lin monthly, Dr. Pinello biweekly, and Dr. Huangfu quarterly. Additionally, Tim, Dr. Pinello, and I will meet every third week alongside a small group of collaborators to discuss progress on **Aim 1** of this grant. I have a longstanding relationship with Dr. Pinello, with 33 co-authored manuscripts and several co-mentored trainees who have gone on to further career success.

Tim will avail himself of the many career development opportunities available to him at BCH, HSPH, Massachusetts General Hospital, and the Broad Institute in pursuit of his training goals. Tim plans to audit CHEM 177, a genome editing class taught by Dr. David Liu, which explores chemistry, biology, and ethical implications, during the first year of the K99 phase of the award. Additionally, Tim plans to take STAT 364, led by his co-mentor, Dr. Xihong Lin, covering modern topics in AI and deep learning from a statistical perspective, in the second year of the K99 phase. Moreover, Tim intends to take the "Techniques to Writing a Competitive Grant" class offered by Harvard Catalyst. Tim will continue to attend key conferences (e.g., RECOMB-seq, American Statistical Association Genetics and Genomics conference, etc.) and local colloquia (e.g., Program in Quantitative Genomics Seminar, Statistical Genetics seminar, etc.) relevant to his research program. Furthermore, Tim will participate in NIH-funded consortia, including the Impact of Genomic Variation on Function (IGVF) consortium and Somatic Cell Genome Editing (SCGE) consortium. Finally, Tim will meet regularly with his mentors, who will provide guidance on his research and evaluate his progress.

In summary, Tim has crafted a comprehensive training plan and forged a highly original research program addressing critical needs in the analysis of CRISPR data. I am fully assured that Tim will be an outstanding faculty member, and I enthusiastically give his application my strongest support.

Sincerely, Daniel E. Bauer, MD, PhD

Xihong Lin
Professor of Biostatistics
Chair and Professor of Statistics
Coordinating Director
HSPH Program in Quantitative Genomics
Harvard University

September 21, 2025

Dear K99/R00 Review Committee,

I am delighted to support Dr. Timothy (Tim) Barry's application for the NIH K99/R00 pathway to independence award. Tim joined my lab in January of 2024 after completing his PhD in Statistics at Carnegie Mellon University. I have been pleased to watch Tim grow as a scientist as he has honed his research program, sharpened his grant-writing skills, and developed proficiency in several new areas relevant to his proposal, including deep learning, causal inference, and CRISPR profiling experiments. As a result of funding complications at Harvard University, Tim transferred to the lab of his co-mentor, Dr. Daniel Bauer, at Boston Children's Hospital in July of 2025. Although Dr. Bauer is now Tim's primary mentor, I remain fully committed to mentoring Tim and supporting his transition to an independent faculty position.

Tim's K99/R00 proposes to develop innovative and much-needed statistical and machine learning methods for the analysis of CRISPR profiling and screening data. CRISPR profiling experiments aim to characterize the editing activity of CRISPR systems at on- and off-target loci. While computational tools (e.g., CRISPResso) are available for characterizing editing events, statistical methods for estimating editing rates and comparing editing activity rates conditions are limited. Tim proposes to develop CRISPR-DE, a comprehensive toolkit for statistically rigorous analysis of CRISPR profiling data. On the CRISPR screen front, Tim proposes to develop DML-Perturb and IV-Perturb, methods that will leverage novel deep learning and instrumental variables frameworks, respectively, to more accurately estimate perturbation effects and detect gene modules in CRISPR screens. I expect Tim's methods to find widespread use among labs conducting CRISPR experiments, accelerate therapeutic development, and elucidate disease mechanisms.

My lab's research focuses on the development of statistical, machine learning, and AI methods for the analysis of massive-scale genetic, genomic, epidemiological, and health data. My lab is a leading lab in developing statistical and ML methods for rare variant association testing in whole-genome sequencing (WGS) analysis. My lab also has done pioneering work in variant function annotation, WGS analysis, integrative analysis of genetic and genomic data, Mendelian randomization, causal mediation analysis, and many other areas. For example, our work integrates multi-faceted variant functional annotations including single cell functional annotations in WGS analysis. I am the PI of the NCI Outstanding Investigator Award (R35, 2015-2029). I also am the MPI of Harvard-UMass Predictive Modeling Center of the NHGRI Impact of Genomic Variation on Function (IGVF) Program.

I am an elected member of the National Academy of Sciences and the National Academy of

Department of Biostatistics
655 Huntington Ave
Building II, Room 419
Boston, MA 02115

TEL 617 432 2914
FAX 617 432 5619
xlin@hsph.harvard.edu

hsph.harvard.edu/xihong-lin

Medicine. Additionally, I am a recipient of the COPSS Presidents' Award and an elected fellow of the American Association for the Advancement of Science, American Statistical Association, and Institute of Mathematical Statistics. I have trained 35 doctoral students and 28 postdoctoral fellows. I have received the mentoring award from Harvard School of Public Health twice. Several of my former and current trainees, including Dr. Xinan Wang and Dr. Haoyu Zhang, have received the K99/R00 Award. Moreover, many of my former postdoctoral research fellows now hold independent faculty positions in government, including at the National Cancer Institute, and in leading academic departments, including at Duke University, Harvard University, University of Pennsylvania, the University of North Carolina at Chapel Hill, and the University of Michigan. I would expect Tim to launch an independent research lab during the R00 phase of this award.

I will work with Tim to help him accomplish his training goals and ultimately achieve independence. I will meet with Tim monthly, providing guidance on all aspects of deep learning (Skill 2) and causal inference (Skill 3) relevant to his proposal. With respect to deep learning, I will mentor Tim on transformer-based architectures for single-cell data, representation learning, large language models, and diffusion models; with respect to causal inference, I will mentor Tim on debiased machine learning, efficient influence function approaches, instrumental variables methods, and Mendelian randomization. Tim plans to audit my class (STAT 364) on statistical deep learning and AI in the second year of the K99 phase of the award. When the time comes for Tim to apply to independent faculty positions, I will help him hone his application materials.

Tim is an exceptional early-career scientist, driven to develop innovative solutions to pressing challenges in CRISPR profiling and screening analysis. It has been a pleasure to watch him take strides toward independence over the past year and a half, and I enthusiastically endorse his application for this prestigious fellowship.

Sincerely,

Xihong Lin, Ph.D.
Professor of Biostatistics
Chair and Professor of Statistics
Coordinating Director, HSPH Program in Quantitative Genomics
Harvard University

Dana-Farber Cancer Institute | Boston Children's
Dana-Farber/Boston Children's Cancer and Blood Disorders Center

HARVARD MEDICAL SCHOOL TEACHING HOSPITAL

David A. Williams, MD
Chief of the Division of Hematology/Oncology, Boston Children's Hospital
President, Dana-Farber/Boston Children's Cancer and Blood Disorders Center
Associate Chairman, Department of Pediatric Oncology, Dana-Farber Cancer Institute
Leland Fikes Professor of Pediatrics, Harvard Medical School

September 17, 2025

Re: PA-24-194, K99/R00 Institutional Support Letter for Dr. Timothy Barry

Members of the Scientific Review Committee:

As Chief of the Division of Hematology/Oncology at Boston Children's Hospital (BCH), I am fully supportive of Dr. Timothy Barry's research and commit to fostering a professional environment to allow him the full opportunity to complete his development into an independent investigator.

Dr. Barry transitioned from the Harvard School of Public Health to BCH in July of 2025 after funding challenges at Harvard led him to seek new a new home institution. This transition has been smooth, as Dr. Barry's research program remains unchanged, and his mentorship team of Dr. Daniel Bauer, Dr. Xihong Lin, Dr. Luca Pinello, and Dr. Danwei Huangfu remains intact and fully committed to his transition to independence. Dr. Barry has demonstrated remarkable resilience and scientific achievement. He has done highly impactful work in single-cell CRISPR screen (perturb-seq) analysis and robust statistics, and the methods that he has developed are used daily by leading genomics labs nationwide. His mentorship team consists of experts in genome editing, statistics, and machine learning and is well-equipped to provide guidance on his career progression.

The institutional environment at BCH is internationally recognized for its research and training programs in therapeutic genome editing, CRISPR screening, single-cell biology, and other topics relevant to Dr. Barry's research and training plans. The Division of Hematology/Oncology at BCH includes the Department of Pediatric Oncology at Dana-Farber Cancer Institute (DFCI), and all staff and faculty carry joint appointments in the Division and the Department of Pediatrics at Harvard Medical School (HMS). Intellectual interactions among our department members are fostered by: 1) Pediatric Hematology/Oncology research seminars, which focus primarily on outside speakers, but also provide 'senior trainees' toward the end of their training an opportunity to present in a formal setting with all Hematology/Oncology faculty present; 2) a weekly 'floor meeting' in which all trainees present ~once/year; this meeting is attended by all BCH faculty focused on hematology and stem cell research; and 3) numerous Harvard Medical area and University seminars and workshops. There are also theme-specific journal clubs that meet regularly.

The Division of Hematology/Oncology prides itself on transitioning some of the nation's most talented hematology and oncology trainees to successful independent careers. This is accomplished through attentive one-on-one mentorship and career guidance, regularly scheduled advisory meetings, weekly departmental floor meetings and research seminars, and a commitment to providing feedback on trainee manuscripts, grants, and oral presentations. Fellows also have access to state-of-the-art computing resources through Harvard Medical School, BCH, Harvard Arts and Sciences, and Massachusetts General Hospital. This environment will continue to allow Dr. Barry to draw on tremendous intellectual and physical resources in a vibrant and collaborative setting.

Should Dr. Barry receive this award, he will devote 100% of his time to the proposed project and will not have any other duties or commitments during the project period. Importantly, our division's commitment to his salary, resources, and professional development is not dependent on the outcome of this application. Dr. Barry is forging a highly original and exciting research program at the intersection of statistics, machine learning, and CRISPR technology. I am personally committed to ensuring his continued development as he transitions into an independent investigator position.

Sincerely yours,

David A. Williams MD

David A. Williams, MD
Chief of the Division of Hematology/Oncology, Boston Children's Hospital

CANDIDATE'S BACKGROUND

I am a statistician working at the intersection of statistical methodology, machine learning, and CRISPR technology. My overarching goal is to develop powerful methods for the robust analysis of CRISPR data. I am especially interested in CRISPR profiling experiments and CRISPR screens. CRISPR profiling experiments aim to characterize the editing activity of therapeutically relevant CRISPR systems at on- and off-target loci, accelerating safe therapeutic development. CRISPR screens, meanwhile, target CRISPR perturbations to genes or noncoding regulatory elements and assess the impact of these perturbations on cellular phenotypes via single-cell sequencing, bulk sequencing, cell sorting, or cell proliferation. My career objective is to build innovative statistical and machine learning methods for CRISPR profiling and screening data analysis, accelerating therapeutic development and improving understanding of gene function and regulation at scale.

Undergraduate. My interest in science was sparked at the University of Maryland at College Park (UMD), where I studied mathematics and computer science, supported by a Howard Hughes Medical Institute Fellowship and a Banneker-Key Scholarship (UMD's most prestigious scholarship). My initial exposure to statistics came through research: I worked as a research assistant in the UMD Ecology Department and at the NIH in nearby Bethesda, MD, tackling problems in statistical ecology, statistical computing, and medical statistics. I found that statistics combined my interest in mathematics and computing with real-world application. My undergraduate research resulted in two first-authored publications and inspired me to pursue graduate study in statistics.

PhD. I completed my PhD in Statistics at Carnegie Mellon University, one of the leading statistics departments nationwide. I focused my PhD research on statistical and computational methods for single-cell CRISPR screen (i.e., perturb-seq) data analysis. Perturb-seq combines CRISPR genome editing and single-cell sequencing to survey the impact of genetic perturbations on individual cells. Perturb-seq has accelerated progress in functional genomics, enabling the mapping of noncoding regulatory elements and variants to target genes and the dissection of gene regulatory networks at scale. Across a series of three sole first authored papers (Barry et al. 2021, *Genome Biology*; Barry et al. 2024, *Genome Biology*; Barry et al. 2024, *Biostatistics*), I developed SCEPTRE, an end-to-end pipeline for statistically rigorous and computationally efficient perturb-seq data analysis. SCEPTRE is supported by an array of novel statistical and computational methods for assigning guide RNAs to cells, testing for association between guide RNA presence or absence and gene expression, and ensuring rigorous quality control. I carefully demonstrated that SCEPTRE improved upon all existing methods, and I also established *why* SCEPTRE enjoyed this performance gain from a statistical perspective. Finally, I wrote a comprehensive e-book – "Hands on single-cell CRISPR screen analysis" – guiding users through the entire process of analyzing their perturb-seq data using SCEPTRE, from importing data to generating figures.

SCEPTRE rapidly has become the de facto method for perturb-seq analysis within the field. SCEPTRE is used routinely by leading functional genomics labs worldwide, including at Stanford University (Engreitz Lab), UC San Francisco (Pollen Lab), Memorial Sloan Kettering (Huangfu Lab), University of Washington (Shendure Lab), Heidelberg University (Velten Lab), and many other institutions. Moreover, SCEPTRE is prominent within the biotechnology sector: it is the first-ever third-party package for perturb-seq analysis endorsed by 10x Genomics (the main commercial supplier of single-cell experimental kits; www.10xgenomics.com/analysis-guides/single-cell-crispr-screen-analysis-with-sceptre), and it has been incorporated into the drug discovery pipeline of CAMP4, a Boston biotechnology startup. I have been active in helping biologists apply SCEPTRE to analyze their data. For example, I collaborated with the Sanaja Lab (NY Genome Center) and the Lappalainen Lab (KRH Royal Institute of Technology) to apply SCEPTRE to map noncoding variants associated with blood traits to their cognate genes; this work, on which I am coauthor, appeared in *Science* (Morris et al. 2023, *Science*). Additionally, I helped integrate SCEPTRE into the official Impact of Genomic Variation on Function (IGVF) CRISPR screen pipeline, where it is being used to process IGVF perturb-seq datasets.

Postdoc. I joined the lab of Dr. Xihong Lin at the Harvard School of Public Health, where I developed the permuted score test, a novel approach to differential analysis based on permuting score test statistics. The permuted score test provably controls false positives across a much broader range of settings than negative binomial regression while nevertheless approximately matching negative binomial regression with respect to power and computational efficiency (Barry et al. 2025, *arXiv preprint*). The permuted score will play a key role in **Aim 1** of my proposed K99/R00 research. Due to unexpected funding challenges at Harvard University, I transitioned into the lab of my co-mentor, Dr. Daniel Bauer, at Boston Children's Hospital in July of 2025. My research program and mentorship team have remained the same (although Dr. Bauer is now my primary mentor).

CAREER GOALS AND OBJECTIVES

My career objective is to launch an independent research lab at a leading institution focused on developing innovative statistical and machine learning methods for the analysis of CRISPR data. CRISPR technology is evolving rapidly: new editors, including base and prime editors, allow for highly precise, single-variant substitutions. Meanwhile, advances in single-cell sequencing (e.g., multimodal sequencing, combinatorial barcoding) are powering a new generation of information-rich, high-resolution CRISPR screens. My future lab will be active in all areas of CRISPR data analysis, from CRISPR profiling analysis to single-cell, flow cytometry, bulk RNA-seq, and cell proliferation CRISPR screen analysis. I have assembled an expert mentorship team to help me gain the skills required to complete this grant, comprising Dr. Daniel Bauer (Boston Children's Hospital), Dr. Xihong Lin (Harvard University), Dr. Luca Pinello (Massachusetts General Hospital), and Dr. Danwei Huangfu (Memorial Sloan Kettering). At the completion of the K99 phase of this award, I will be well-positioned to launch an independent research lab at the intersection of CRISPR technology, statistics, and machine learning.

Justification for award and short-term goals. Through my PhD and postdoctoral work, I have gained expertise in several fields, including perturb-seq analysis and robust statistics. However, to successfully complete the aims of this K99/R00, I require training in three additional areas: CRISPR editing and screening technology (**Skill 1**), deep learning methodology (**Skill 2**), and statistical causal inference methodology (**Skill 3**). These three skills are closely connected to the aims of my grant.

Aim 1 (K99 phase) proposes to develop *CRISPR-DE*, a comprehensive toolkit for the statistical analysis of CRISPR profiling experiments. To accomplish this aim, I must expand my understanding of the biology of CRISPR editors, including Cas9, base, and prime editors (**Skill 1**). For instance, Cas9, base, and prime editors exhibit different mutation signatures, and I may need to account for these differences in developing statistical models for CRISPR profiling data. **Aim 2** (K99/R00 phase) proposes to develop *DML-Perturb*, a novel deep learning framework for transcriptome-wide perturb-seq analysis. Completing this aim will require me to gain proficiency in several areas: recent advances in perturb-seq assay design (e.g., dual guide RNA lentiviral delivery systems) (**Skill 1**), state-of-the-art deep learning methods for single-cell data (e.g., transformer-based single-cell neural network architectures) (**Skill 2**), and statistical techniques for estimating causal effects using machine learning algorithms (e.g., double machine learning and efficient influence function approaches) (**Skill 3**). Finally, **Aim 3** (R00 phase) is centered on developing *IV-Perturb*, a statistical method to account for variable perturbation effectiveness in CRISPR screen analysis. For this aim, I intend to deepen my knowledge of instrumental variables methods in causal inference, including structural equation models and structural mean models (**Skill 3**). Additionally, I plan to expand my expertise in CRISPR screening assays beyond perturb-seq (**Skill 1**), including arrayed bulk RNA-seq and flow cytometry screens.

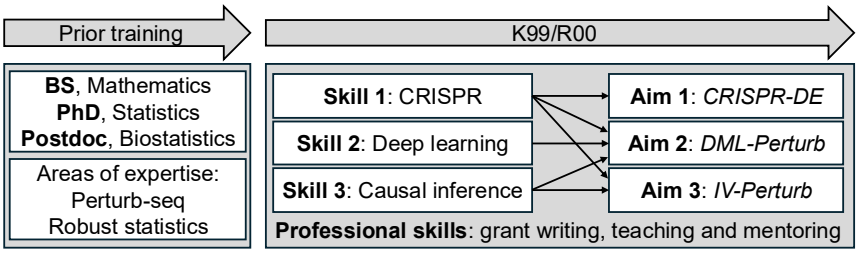

***Figure 1**: Career trajectory. In the "K99/R00" panel, an arrow points from a given skill to a given Aim if that skill is required to complete that Aim.*

In addition to the technical skills identified above, I seek to cultivate several professional skills during the K99 phase, including grant writing and teaching/mentoring skills. Cultivating these skills will enable me to provide my trainees with funding and effective mentorship upon launching my own lab.

How I will differentiate from my mentor. My primary mentor, Dr. Daniel Bauer, is an expert in therapeutic genome editing, CRISPR screens, functional genomics, and blood diseases. Dr. Bauer identified the enhancer region and guide RNA used to disrupt this region in the first-ever FDA-approved gene editing therapeutic, namely Casgevy (for sickle-cell disease and beta thalassemia). Additionally, Dr. Bauer, in collaboration with my co-mentor, Dr. Pinello, has developed widely used computational methods for analyzing CRISPR editing and screening data, including CRISPResso and CRISPR Bean. I will distinguish myself from my mentor by delving deeply into the statistical aspects of CRISPR data analysis, bringing to bear sophisticated and robust data analytic frameworks. My lab will stay abreast of the latest developments in both statistical methodology and CRISPR profiling and screening analysis, leveraging the former to develop state-of-the-art tools for the latter.

Long term goals. I plan to apply for an independent faculty position at a leading institution within 1.5 years of starting the K99 phase of this award. To ensure that my lab is well-funded and capable of recruiting talented PhD students and postdocs, I will apply for an R35 (or similar) during the second year and an R01 (or similar) during the third year of my independent faculty position. I will foster collaborations with experimental biologists, building on my relationships with the many researchers worldwide who already use SCEPTRE to analyze their

perturb-seq data. I will mentor successive waves of PhD students and postdocs, who will go on to initiate their own independent research programs in statistics, computer science, genomics, and genetics.

CAREER DEVELOPMENT/TRAINING ACTIVITIES DURING AWARD PERIOD

Primary mentorship. Dr. Daniel Bauer is an Associate Professor at Boston Children's Hospital and an international leader in therapeutic genome editing, CRISPR screening, blood trait GWAS, and blood diseases. Dr. Bauer will mentor me in all aspects of CRISPR biology and technology (**Skill 1**), including CRISPR editing mechanisms (e.g., sequence recognition, DNA repair pathways, epigenetic remodeling, etc.), CRISPR screen design and analysis (e.g., pooled and arrayed screens, tiling screens, molecular phenotyping, etc.), and off-target analysis (e.g., amplicon sequencing, DNA tagging sequencing, etc.). I will present my research in Dr. Bauer's lab meeting (Tuesdays, 9:30-11am) twice per year and continue to meet with him biweekly (Thursdays, 2-3pm). Dr. Baur's office is adjacent to the lab space in which I work, facilitating frequent in-person interactions.

Mentor	Role	Institution	Meeting frequency	CRISPR	Deep learning	Causal
Dr. Bauer	Primary	Boston Children's	Biweekly	✓		
Dr. Lin	Co-	Harvard Public Health	Monthly		✓	✓
Dr. Pinello	Co-	Massachusetts General	Weekly	✓	✓	
Dr. Huangfu	Co-	Memorial Sloan	Quarterly	✓		

***Table 1**: My mentorship team. The columns in grey indicate which mentor will train me in which skill.*

Co-mentorship. I have assembled an expert co-mentorship team comprising Dr. Xihong Lin, Dr. Luca Pinello, and Dr. Danwei Huangfu. Dr. Lin is an international leader in statistical methods for genetics and genomics. Dr. Lin developed the SKAT method, a widely used (>2,800 citations) tool for rare variant association testing in WGS analysis. Additionally, Dr. Lin has extensive experience in causal inference methods for genetics, having developed mediation analysis and instrumental variables (i.e., Mendelian randomization) approaches for integrating data from genomic, epigenomic, and proteomic modalities. Moreover, Dr. Lin has expertise in deep learning and AI methods for variant function annotation, polygenic risk prediction, and other common genetics tasks. Dr. Lin will train me in deep learning (**Skill 2**) and causal inference (**Skill 3**) techniques. Additionally, an expert in GWAS, Dr. Lin will mentor me in the interpretation of variant-targeting CRISPR screens. I will meet with Dr. Lin quarterly. Her office is a five-minute walk from my lab space at Boston Children's. Dr. Pinello is an Associate Professor at Massachusetts General Hospital (MGH) and one of the world's foremost leaders in computational methods for CRISPR data. Dr. Pinello has created essential tools for CRISPR profiling and screening analysis, including CRISPResso (>1,800 citations). Additionally, Dr. Pinello has built popular methods for single-cell data analysis, including SIMBA, a graph neural network approach for generating cell and feature embeddings. Dr. Pinello will train me in topics related to CRISPR editing and screening analysis (e.g., modeling CRISPR profiling data, aligning and counting edited/unedited reads; **Skill 1**) and deep learning techniques for single-cell data (**Skill 2**). I will meet one-on-one in-person with Dr. Pinello weekly (Fridays, 4:30-5pm) and present twice per year each at his weekly journal club (Wednesdays, 1:00-2:00pm) and lab meeting (Fridays, 9-11am). I have badge access to MGH and a dedicated desk in Dr. Pinello's lab space, a 30-minute bike ride from the School of Public Health. Dr. Danwei Huangfu is a Professor of Cell and Developmental Biology at Memorial Sloan Kettering. Dr. Huangfu's Lab uses genome-scale CRISPR screening approaches to discover enhancers and identify key genetic and epigenetic regulators of development. Additionally, Dr. Huangfu's Lab is one of the core labs generating perturb-seq data as part of the NIH IGVF Consortium. I have collaborated with the Huangfu Lab to apply SCEPTRE, my perturb-seq method, to analyze their in-house perturb-seq data. Building on this track record, I will collaborate with the Huangfu lab in applying *DML-Perturb* (**Aim 2**) and *IV-Perturb* (**Aim 3**) to analyze their CRISPR screen data. Moreover, Dr. Huangfu will train me in key aspects of CRISPR biology relevant to my proposal, including emerging CRISPR editing modalities, screening techniques, and delivery methods (**Skill 1**). Dr. Huangfu and I will meet one-on-one quarterly over Zoom and interact at IGVF meetings.

Evaluation. My mentors will evaluate my performance once per year during the K99 phase of this award. They will assess my research productivity (e.g., manuscript preparation, presentations at conferences, software development), progress toward training goals (e.g., completion of courses and workshops) and overall trajectory toward independence. I will update my training plan as necessary in accordance with their feedback.

Courses. I will audit one science course per year during the award period to improve my technical skills. I plan to take "The Chemistry, Biology, and Societal Implications of Genome Editing" (CHEM 177) during the first year of the award period (Fridays, 3-5:45 pm). This course is taught by Dr. David Liu, world-renowned gene editing researcher and collaborator of Dr. Bauer and Dr. Pinello. CHEM 177 covers the fundamentals of Cas9, base, and prime editing, as well as therapeutic genome editing and ethics considerations (**Skill 1**). Next, I plan to take

"Scalable Statistical and Machine Learning Methods" (STAT 364) during the second year of the award period (Mondays, 10:30-11:45am). This course, taught by my co-mentor Dr. Lin, covers key topics in deep learning and AI methods for genomics, including model architectures, transfer learning, and representation learning (**Skill 2**).

Grant writing. I have twice participated in the Harvard School of Public Health K99/R00 workshop. I plan to further improve my grant writing skills by taking the "Techniques to Writing a competitive grant" short course offered by Harvard Catalyst. This course provides instruction on concrete strategies for writing fundable grants and will position me for success in writing an R35 and R01 grant during the R00 phase of this award.

Consortium work. I am a member of the CRISPR focus group (FG) of the NIH IGVF Consortium. The CRISPR-FG leverages CRISPR screens (including perturb-seq) to understand the function of GWAS variants. I helped incorporate SCEPTRE into the data processing pipeline of the CRISPR-FG, where it is now used to process perturb-seq datasets generated throughout the IGVF. Additionally, I was a leader of the 2025 CRISPR-FG jamboree, where I headed the effort to define metrics of guide RNA assignment quality and compare competing guide RNA assignment methods. I attended the second and third annual IGVF meetings (in St. Louis in 2023 and Seattle in 2024, respectively) and frequently attend the biweekly online CRISPR-FG meetings. I will continue to participate in the IGVF during the award period, helping organize future jamborees and coauthor publications headed by the CRISPR-FG. Recently, I have begun to participate in the NIH Somatic Cell Genome Editing (SCGE) consortium, and I plan to attend SCGE meetings during the award period.

Mentorship activities	K-1	K-2	R-1	R-2	R-3
Regular meetings with mentors	✓	✓			
Timeline for Aims					
CRISPR-DE (Aim 1)	✓	✓			
DML-Perturb (Aim 2)		✓	✓	✓	
IV-Perturb (Aim 3)				✓	✓
Technical courses					
Genome editing (CHEM 177, Dr. Liu)	✓				
Deep learning and AI (STAT 364, Dr. Lin)		✓			
Grant writing and job search					
Harvard Catalyst grant writing course		✓			
HSPH faculty job market workshop		✓			
Submit grant (R35 → R01)				✓	✓
Conferences, colloquia, consortia					
Conference travel (3x/year)	✓	✓	✓	✓	✓
Local colloquia (as needed)	✓	✓			
IGVF and SCGE participation	✓	✓			
NHGRI annual trainee meeting	✓				

***Table 2**: Timeline of career development activities. "K-1": first year of K99 phase, etc.*

Conferences and colloquia. I have presented at eight conferences since starting my postdoc (two as an invited speaker), and I plan to continue attending conferences during the award period. I will submit abstracts to the CSHL Genome Engineering Conference, IGVF annual meeting, SCGE annual meeting, American Statistical Association Genetics and Genomics Conference, and RECOMB-seq. Moreover, I will attend the following Harvard School of Public Health and Broad Institute colloquia on an as-needed basis: Program in Quantitative Genomics seminar, Statistical Genetics seminar, and Model, Inference, and Algorithms seminar. These seminars frequently feature speakers specializing in causal inference and deep learning methods, and I have benefited and will continue to benefit from discussing research trends and recent publications with speakers during question-and-answer sessions and informally over lunch (**Skills 2** and **3**).

Mentoring and teaching. I closely mentored a master's student over the course of a year during my PhD, which helped me gain valuable mentorship experience. I plan to mentor a student from Dr. Pinello's group during the award period. I already have begun working with a visiting undergraduate student from Cambridge University, Mr. Benjamin Vyshedskiy. I mentored Mr. Vyshedskiy on statistical analysis of CRISPR profiling data; a manuscript communicating this work, on which Mr. Vyshedskiy and I are coauthors, is forthcoming. I will mentor either Mr. Vyshedskiy or another student in the Pinello Lab in completing a component of **Aim 1** of this grant. Additionally, I will gain teaching experience by giving a guest lecture in Dr. Pinello's annual "Introduction to Computational Biology and Bioinformatics" class (BST 282), a master's level class for biostatistics students.

Job search. I will apply for an independent faculty position within 1.5 years of beginning the K99 phase of this award. I will solicit feedback from my mentors on my application materials, including my research statement, teaching statement, and job talk. I will request that my mentors introduce me to their professional network so that hiring committees will be familiar with my work. Moreover, I will take the two-session academic job market application workshop offered by the School of Public Health to enhance my application materials.

TRAINING IN THE RESPONSIBLE CONDUCT OF RESEARCH

I will complete Responsible Conduct of Research (RCR) course offered by Boston Children's Hospital in Fall of 2026 (Tuesdays, 10-11am). This course fulfills the NIH requirement for formal instruction in RCR for NIH-supported trainees.

1. Format of Instruction: The course consists of a series of in-person, instructor-led, interactive lectures. Lectures include structured discussions and analysis of case studies.

2. Subject Matter: The course covers a broad spectrum of ethical and regulatory issues in research, including the following:
- Responsible authorship
- Collaborative science
- Mentor-mentee relationships
- Research misconduct
- Data management
- Peer review
- Contemporary ethics
- Conflicts of interest

3. Faculty Participation: The course is taught by Judy Fleming (PhD) of the Office of Research Affairs and Integrity at Boston Children's Hospital.

4. Duration of Instruction: The course runs over eight weeks during the semester, with weekly 60-minute sessions (totaling 8 contact hours). Participants are required to attend all sessions to pass the course. Additionally, participants are required to complete CITI training modules and write a statement demonstrating understanding of course material.

5. Frequency of Instruction: Pursuant to NIH requirements, I will take the RCR course once during my postdoctoral research training and once every four years thereafter.

FACILITIES AND OTHER RESOURCES

Workspace:
The Division of Hematology/Oncology occupies space on the eighth floor of the Karp Research Building at Boston Children's Hospital, where Dr. Barry works in a shared laboratory setting. The department offers multiple conference rooms fully equipped with teleconferencing systems, projection equipment, and wall-to-wall whiteboards to facilitate collaborative meetings and seminars. The Bauer Lab employs a dedicated lab manager who oversees administrative responsibilities, including travel arrangements, scheduling, and publication fee processing. Additionally, Dr. Barry receives support from two grant managers assigned by the Division.

Computing:
Dr. Barry has access to two primary high-performance computing (HPC) resources: the Harvard FASRC Cannon cluster (via Dr. Pinello's appointment in the Department of Biostatistics at the Harvard T.H. Chan School of Public Health) and the Harvard Medical School (HMS) O2 cluster (via Dr. Bauer's appointment at Harvard Medical School).

The FASRC Cannon cluster is a large-scale, liquid-cooled computing system supporting intensive scientific workloads. Its infrastructure includes Lenovo NeXtScale servers with Intel Cascade Lake and Sapphire Lake processors, a high-speed 400 Gbps InfiniBand interconnect, and a growing GPU partition (connected at 100 Gbps). Over 60 petabytes of high-performance storage -- managed via Isilon, Lustre, CEPH, and NFS filesystems -- enables both high-throughput analysis and secure long-term data storage. The cluster is managed by a dedicated support team and offers reliable, scalable, and secure remote access.

The HMS O2 cluster comprises 390 nodes with a total of over 12,000 CPU cores and approximately 106 terabytes of RAM. It features diverse node types, including high-memory and GPU-enabled nodes, linked via Ethernet and InfiniBand interconnects. O2 offers a comprehensive array of software packages for bioinformatics and computational genomics applications. Together, the Cannon and O2 clusters will facilitate the data-intensive analyses required of Dr. Barry's project.

Library resources:
Through his Harvard Sponsored Role, Dr. Barry has unrestricted access to Harvard University's vast library network. The main cataloging tool is HOLLIS, a unified search platform providing access to millions of books, articles, and digital resources. Any digital material unavailable through HOLLIS can be accessed via Harvard Library's interlibrary loan program. Additionally, the Karp Research Building (where Dr. Barry works) is a five-minute walk from the Francis A. Countway Medical Library, which contains 480,000 volumes and 5,000 periodicals, to which Dr. Barry has access.

Proximity to mentors
Dr. Barry works in close physical proximity to his mentors and co-mentors. Dr. Bauer's office in the Division of Hematology/Oncology is down the hall from the lab space in which Dr. Barry works, facilitating frequent in-person interactions. Dr. Lin's office is in the Department of Biostatistics at the Harvard T.H. Chan School of Public Health, a five-minute walk from the Karp Research Building. Dr. Pinello's lab is located at Massachusetts General Hospital (MGH) Navy Yard, a 30-minute bike ride from the Karp Research Building. Dr. Barry has physical access to MGH and has a dedicated desk in Dr. Pinello's lab. Dr. Barry also is in close physical proximity to other faculty members in the Longwood Medical Area whose expertise may be of relevance to his research, including Dr. Rong Ma (statistical methods for single-cell genomics), Dr. Alkes Price (statistical methods for genetics and functional genomics), and Dr. Mark Daly (computational methods for genetics) (see *Description of Institutional Environment*).

DESCRIPTION OF INSTITUTIONAL ENVIRONMENT

My primary institution is Boston Children's Hospital (Division of Hematology and Oncology). Through my co-mentors Dr. Xihong Lin (Harvard School of Public Health) and Dr. Luca Pinello (Massachusetts General Hospital and Harvard School of Public Health), I have an appointment at the Harvard School of Public Health (Harvard Sponsored Role). Additionally, through Dr. Pinello, I have an appointment at Massachusetts General Hospital. Finally, I am an affiliate member of the Broad Institute. The institutions at which I have an affiliation -- Boston Children's Hospital, Harvard School of Public Health, Massachusetts General Hospital, and the Broad Institute -- are all located in or around downtown Boston and are within a thirty-minute bike ride of one another. This network of institutions provides a rich environment in which to carry out my training and career development activities.

Faculty members. Many leading researchers in CRISPR biology and statistical genetics and genomics work at Boston Children's Hospital, the Harvard School of Public Health, Massachusetts General Hospital, and the Broad Institute. The office of my primary mentor, Dr. Daniel Bauer, is adjacent to the lab space in which I work, facilitating frequent in-person interactions. Additionally, Dr. Xihong Lin's lab is housed in the Department of Biostatistics at the Harvard School of Public Health, a five-minute walk from Boston Children's Hospital. Finally, Dr. Luca Pinello has a lab space at Massachusetts General Hospital Navy Yard, a thirty-minute bike ride from Boston Children's Hospital. I have a dedicated desk in Dr. Pinello's lab space, located down the hall from Dr. Pinello's office.

I am physically colocalized with a variety of other faculty members working on topics relevant to my training plan and research objectives. For example, Dr. Alkes Price is a leading statistical geneticist who leverages functional genomics data to understand the etiology of disease-associated variants; his lab is in the Department of Epidemiology at the Harvard School of Public Health. Additionally, Dr. David Liu is an international expert in genome engineering, having developed base and prime editing. His lab at the Broad Institute is a thirty-minute bike ride from Boston Children's Hospital. Other key faculty members include Dr. Po-Ru Loh (Harvard Medical Sciences), Dr. Rong Ma (Harvard School of Public Health), and Dr. Mark Daly (Broad Institute).

Career development resources for postdocs. Boston Children's Hospital, Harvard School of Public Health, and Massachusetts General Hospital offer a bevy of career development opportunities for postdocs. I have twice attended the Harvard School of Public Health K99/R00 workshop, which has helped me hone my grant writing skills. When the time comes to apply for faculty jobs, I will participate in the Harvard School of Public Health workshop on creating an engaging faculty job application package. Other postdoc workshops at Boston Children's Hospital, Harvard School of Public Health, and Massachusetts General Hospital cover topics such as writing a compelling Specific Aims statement, designing courses, and building a research portfolio.

Computing and library resources. I have access to state-of-the-art computing and library resources through Boston Children's Hospital, Harvard School of Public Health, and Massachusetts General Hospital (see *Facilities and Other Resources*). The Harvard library network is extensive and provides digital access to every journal required of my research via the HOLLIS catalogue and interlibrary loan program.

Courses. Harvard University and Harvard Medical School offer several classes that are relevant to my training objectives and taught by internationally renowned faculty. I plan to take David Liu's CHEM 177 (the biology of CRISPR editors; class size: <30) and Xihong Lin's STAT 364 (deep learning and AI methods in genomics) during the award period. Additionally, through Harvard Catalyst, I intend to take a course on writing an effective R01 application, which is free to all Harvard affiliates.

Seminars and colloquia. Harvard School of Public Health, Massachusetts General Hospital, Boston Children's Hospital, and the Broad Institute offer an array of colloquia relevant to my interests, including the Program in Quantitative Genomics (PQG) seminar, the Statistical Genetics seminar, Department of Biostatistics Colloquium series, and the Models, Inference and Algorithm series. Speakers from around the country visit to speak at these colloquia, with topics covering genome editing, causal inference, single-cell genomics, deep learning, and AI.

Greater Boston. The greater Boston area is home to myriad biotechnology companies, many of which were spun out of research labs at Harvard and affiliated hospitals. I have connected with scientists in industry to learn about how genomics is being applied in practice. For example, I had lunch with a computational biologist from CAMP4, a Cambridge-based biotechnology startup that is using my SCEPTRE tool as part of their drug discovery pipeline.

Budget Justification

Mentored K99 Phase Years 1 & 2

Personnel:

Timothy Barry, Postdoctoral Research Fellow at Boston Children's Hospital. Dr. Barry is the PI for this Pathway to Independence Award and will devote 12 calendar months effort to the project. Dr. Barry will be responsible for developing statistical and machine learning methods for analyzing CRISPR screening and profiling data. A salary of $80,358 is requested for the first year and $83,572 for the second year of the R00 phase. Dr. Daniel Bauer will serve as the mentor on this project during the K99 phase.

Other Direct Costs:

Domestic travel: ($4,000 per year, Years 1-2): Funds are requested for Dr. Barry to attend two domestic conferences or workshops per year. Conferences will include the Joint Statistics Meetings, the International Biometric Society ENAR meeting, RECOMB-seq, the American Society of Human Genetics (ASHG) Conference, the Impact of Genomic Variation on Function Consortium Annual Meeting, and the Biology of Genomes Conference at Cold Spring Harbor. This allocation also will be used to attend the annual IGVF trainee meeting. The amount includes travel, airfare, ground transportation, accommodation, registration fees, and meals.

Foreign Travel: ($5,000 per year, Years 1-2): Funds are requested for Dr. Barry to attend two international conferences per year. Conferences include the European Society of Human Genetics Conference and the RECOMB Conference (which is held outside the USA every other year). The amount includes travel, airfare, ground transportation, registration fees, and meals.

Claude Code: ($1,200 per year, Years 1-2): Funds are requested for a Claude Code subscription ($100/month, $1,200 per year, $2,400 across Years 1 and 2). Claude Code is an AI coding assistant that will enable Dr. Barry to write code for this project more quickly and accurately.

Laptop: ($2,000 per year, Year 1): Funds are requested to purchase a Macbook Pro 10-Core M4 CPU laptop with 16GB unified memory and 1TB SSD Storage.

Publications: ($3,000 per year, Years 1-2) Funds are requested for the costs of documenting, preparing, publishing, or otherwise making available to others the findings and products of the work conducted under the award.

Indirect costs: Indirect costs are calculated at 8% of the Modified Total Direct Cost base.

R00 Phase (Years 3-5) A detailed budget for years 3-5 of this proposal will be submitted as a part of the requisite K99/R00 transition application when the PI secures an independent faculty position and transitions to new institution. Other Direct Costs add a line item titled 'R00 Independent Phase' and provide the total request for that period (up to $249,000).